Examiner only

4. One type of petrol car costs £12 500 to buy.
The equivalent model as an electric car costs £24 500.

(a) The electric car travels 240 km on a full charge.
It takes 8 hours to fully charge the electric car battery.
A home charging point is rated at 7 kW.
Homes are charged 30 p for each kWh of electricity used.

Use the information above and equations from page 2 to calculate the cost, **in £**, to travel 240 km. [3]

charging cost for 240 km = £ ....

(b) Fuel consumption for the petrol car is 15 km/l (kilometres per litre).

The cost of petrol is £1.60 per litre.

(i) Calculate the fuel cost if the petrol car is driven 240 km. [2]

fuel cost for 240 km = £ ....

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(ii) Both cars are driven 14 400 km per year.

I. Calculate the difference in running costs for one year. [2]

difference in running costs per year = £ ....

II. Calculate the payback time of the **extra cost** if the electric car is bought instead of the petrol car. [2]

payback time = ... years

(c) It is often claimed that electric cars are environmentally friendly because they do not produce greenhouse gases when used.
Explain whether you agree. [2]

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